

LEISHKA PAGAN

Microsoft AI Innovation Challenge – 3rd Place (2026)

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PROFESSIONAL SUMMARY

Security & Cloud Infrastructure Engineer specializing in AI security and detection engineering. 9+ years in high-availability enterprise environments, managing identity for **5,400+ users** as final escalation point. Builds production AI security systems on Azure with code-enforced safety layers. **3rd Place, Microsoft AI Innovation Challenge 2026.**

PROFESSIONAL EXPERIENCE

Security Consultant & Lead Engineer | Women in Cloud

May 2026 – Present Remote

- Lead security and full-stack engineer for a pre-launch platform, owning threat modeling, authentication, authorization, and data handling across backend and frontend, with security designed in from the first commit.
- Hardening the platform against injection attacks, broken access control, dependency risk, and abuse pathways; securing AI-assisted features against prompt injection, model abuse, and adversarial inputs.
- Drive the security and scaling roadmap from internal pilot through controlled expansion to public launch; report to WIC leadership on platform architecture. Product under NDA until launch.

Independent Projects & Applied Research | AI Security & Detection Engineering

2023 – Present

AI Security Systems

- **Placed 3rd internationally** in the Microsoft AI Innovation Challenge (March 2026), an invitation-only competition, as Primary Engineer on ClearStep, an AI decision-support application deployed across 10 Azure services.
- Hardened ClearStep against **14+ attack vectors** with a three-layer safety pipeline (Azure AI Content Safety, Azure OpenAI signal classification, Anthropic Claude reasoning) and a Python validation layer enforcing prompt-injection mitigation and deterministic schema output.
Code: <https://github.com/leishka-pagan/ClearStep>
- Built **Sentinel**, a custom AI-powered vulnerability intelligence platform using a multi-agent architecture with orchestrated reasoning and validation layers, a formal finding state machine, OWASP ASVS/WSTG and MITRE ATT&CK mapping.
- Enforced a hard-scoped safety layer with zero destructive actions; produced evidence-backed JSON and Markdown reports with confirmed, refuted, and inconclusive classifications.
Code: <https://github.com/leishka-pagan/Sentinel>

Security Operations & Detection Engineering

- Authored custom Snort IDS rules from live packet analysis, isolating malicious RDP traffic from baseline behavior with a low false-positive rate.
- Built Splunk detection workflows using custom SPL queries, correlating logs across sources and reducing **3,000+ events** to targeted IOCs, with all confirmed findings mapped to MITRE ATT&CK techniques.
- Conducted Windows event log threat hunting with Hayabusa, authored Sigma rules for cross-platform detection, and recovered credentials from memory using Volatility.

Incident Response & Investigations

- Performed static malware analysis and reverse engineering on live samples, authoring YARA detection rules validated against multi-engine analysis.
- Conducted IR investigations spanning memory forensics for credential recovery, SIEM pivot analysis from exposed POP3 services, and malware identification where AV failed detection. Externally validated through IronCircle TDX Arena (IR Expert + CyberAdvantage certifications).

Infrastructure & Operations Support (Senior Office Specialist) | Metro Net Fire Authority | 911 Dispatch

Anaheim, CA

Mar 2024 – Mar 2025

- Re-engineered the QA reporting workflow, building a VBA automation pipeline executing **240+ discrete actions** per cycle and reducing manual processing time by **75-80%**.

- Extended automation end-to-end: data extraction from individual review workbooks, dictionary-based team assignment across 20+ dispatchers, auto-generated pivot analysis with conditional formatting, and Outlook-integrated PDF distribution.
- Maintained data integrity in the Computer-Aided Dispatch (CAD) system, updating address records, coordinates, and response codes that drive routing accuracy.
- Extracted dispatch logs, call recordings, and radio communications for subpoenas, warrants, and records requests under CJIS-aligned workflows, preserving evidentiary integrity and audit readiness.

Career Break (Family Care + Transition to Cybersecurity Focus)

2020 – 2023

Full-time caregiver during COVID-19; returned to technical practice in 2023 through labs, DFIR, and detection engineering projects.

Systems Administrator, Enterprise Infrastructure & Identity | Spectrum | Time Warner Cable

Jan 2014 – Feb 2020

El Segundo, CA | Formerly Time Warner Cable; retained through acquisition by Charter Communications (2016) | Promoted from Dispatcher II (Feb 2012 – Jan 2014)

- Advanced from sole administrator supporting **100+ dispatch agents** to co-owning identity and access governance for **250–300 users** across multiple facilities; one of two senior administrators on a 4-person team.
- Administered identity and access lifecycle for **5,400+ field technicians** across enterprise platforms (ARRIS, CSG, SageQuest, Kronos, Avaya, Microsoft 365), enforcing RBAC and least-privilege access in SSO-enabled environments.
- Served as the final escalation point for enterprise infrastructure, authentication, and workflow failures across regional dispatch operations, handling **20-40+ escalations daily (spiking post-maintenance)**; resolved issues no other tier could, coordinating directly with ARRIS senior engineering.
- Administered Active Directory identity lifecycle including user provisioning, GPO management, access governance enforcement, and authentication troubleshooting; executed proactive monitoring and remediation initiatives that reduced user-reported incidents by **~15%**.
- Conducted monthly access audits and onboarding/offboarding-triggered reviews, validating permissions across enterprise systems with separate authentication environments to eliminate residual or misconfigured access.
- Led infrastructure upgrades, patch lifecycle management, disaster recovery planning, and overnight maintenance windows; drove enterprise technology transitions including migration to iPad-based field workflows.
- Developed SOPs, escalation playbooks, and visual job aids adopted across administrator and technician teams; led training for dispatch teams and supervisors, reducing recurring support requests and improving onboarding consistency.

TECHNICAL SKILLS

- **AI Security & LLM Engineering:** Prompt injection mitigation (XML delimiting), multi-layer safety pipelines, content safety / guardrails, structured output validation, multi-agent architecture, Anthropic Claude API, Azure OpenAI, OWASP ASVS/WSTG, MITRE ATT&CK mapping.
- **Cloud & Identity:** Azure (App Service, Entra ID, Key Vault, Managed Identities, PIM, Conditional Access, AI Content Safety, Prompt Shields, AI Language, AI Speech, OpenAI, Blob Storage, Cosmos DB, Application Insights), Microsoft Foundry, AWS IAM, Active Directory, Microsoft 365, RBAC, MFA, SSO, Identity Governance, RADIUS/TACACS+, 802.1X, GPO, DefaultAzureCredential
- **Detection & SIEM:** Splunk (custom SPL authoring), Sigma rules, Snort (custom rule authoring from packet analysis), Suricata, Zeek, Hayabusa, Chainsaw, DeepBlueCLI, IDS/IPS detection engineering, detection rule tuning
- **DFIR & Forensics:** Volatility3, Autopsy, KAPE, Velociraptor, memory forensics, timeline analysis, IOC extraction, cryptographic hash verification, evidence integrity, UTF-16 / Base64 payload analysis
- **Malware Analysis & RE:** Ghidra, x64dbg, PeStudio, YARA, ClamAV, VirusTotal multi-engine analysis, static analysis
- **Network & Infrastructure:** LAN/WAN, VLAN segmentation, pfSense, firewall policy, network segmentation, patch management, system hardening, Wireshark, tcpdump, Nmap, Samba/SMB, POP3/SMTP, RDP, DNS, TCP/IP analysis, VPN (IPSec), Zero Trust
- **Development:** Python (Flask, Pydantic), PowerShell, Bash, JavaScript, HTML/CSS, SQL (SQLite), Excel VBA, REST API design, CI/CD (GitHub Actions), Git, Unicorn, Docker
- **Vulnerability Assessment:** Burp Suite, Metasploit, OSINT, Nmap, authorized VDP research (Bugcrowd)

CERTIFICATIONS & EDUCATION

- CompTIA Security+ (SY0-701), Active 2026-2029
- Cybersecurity Professional Certificate, California State University, Long Beach (ThriveDX), coursework across DFIR, threat hunting, network security, Microsoft security administration, and ethical hacking; **150+ hands-on labs completed**
- WiCyS Member; Authorized VDP Researcher (via Bugcrowd)